

Ethan Xing

226-581-3230 | e5xing@uwaterloo.ca | [linkedin.com/ethanxinguw](https://www.linkedin.com/in/ethanxinguw) | github.com/gnixnahte | ethanxing.com

EDUCATION

University of Waterloo, Mechatronics Engineering - Bachelor of Applied Science

GPA: 3.95

Relevant Coursework: Data Structures & Algorithms, Circuits, Digital Computation, Linear Algebra, Calculus I & II

SKILLS

Programming Languages: Python, C/C++, JavaScript, TypeScript, PostgreSQL/SQL, HTML/CSS, Bash

Frameworks & Libraries: React, Django, Three.js, Next.js, Node.js, Tailwind CSS, NumPy, Pandas, LangGraph, LangChain

Tools & Platforms: Git, GitHub, Docker, Supabase, Google Cloud Platform, Codex, Claude Code, mitmproxy, VS Code, n8n

EXPERIENCE

AI Software Engineer

May 2026 – Aug 2026

GrazeAI | Python, JavaScript, Django, Claude Code, Codex, n8n

San Francisco, CA

- Shipped production automation, data, marketing and sales operations platforms, surfaced 10k+ new leads
- Integrated and optimized external APIs and services for downstream services, eliminated 5+ bugs tied to rate limiting
- Optimized webscraping workflows with custom Apify actors, reduced costs by 65% + extended token usage by 25%
- Built LLM-powered lead intelligence pipeline to enrich conference prospects, increasing data coverage up to 21%
- Led CRM cleanup and migration for 30k+ leads and 250k+ emails, improved downstream accuracy by 64%

Robotics Software Engineer and Researcher

Sep 2025 – Aug 2026

University of Waterloo - Robohub | Python, ROS2, MeshCat, Matplotlib, Docker, 3D Bioprinting

Waterloo, ON

- Developed control software system using Python + ROS2 to control a 6-DOF robotic arm with millimeter precision
- Used 3D MeshCat visualizations to optimize and plan toolpaths efficiently, reducing operating costs by 32%
- Benchmarked and conducted performance profiling (found and resolved bottlenecks improving accuracy by 50%)
- Finetuned trajectory planning with validation test cases covering 25+ scenarios, caught 7 bugs to boost reliability
- Wrote technical documentation for engineering team onboarding and common troubleshooting procedures

Executive Director

Sep 2023 – Jun 2024

University of Waterloo - Canadian Youth STEM Conference

Waterloo, ON

- Led logistics for national STEM conference with 300+ attendees featuring Nobel Laureate Donna Strickland
- Creative Director for Innovation Challenge, designing prompt, rules, judging criteria, and team-based challenge structure

PROJECTS

Melodica | *Tone.js, Web Audio API, TypeScript, Node.js, Next.js, Supabase, Google Cloud Platform*



- Built a browser-based digital audio workstation (DAW) with an interactive piano-roll interface for music production
- Implemented a real-time audio scheduling system using Tone.js and the Web Audio API to ensure precise playback
- Optimized frontend performance by implementing viewport-based loading to reduce DOM overhead in the editor
- Designed a scalable JSON-based data model to enable efficient save/load, state management, and audio export to MP3

Scribble | *Python, TensorFlow, Keras, Gemini API*



- Built and deployed an ML-powered full-stack accessibility tool to transcribe handwriting for those with Parkinson's
- Designed a data processing and inference pipeline on 7,000+ handwriting samples with preprocessing and batching
- Achieved 92% accuracy through debugging, batching, pipeline optimization, and iterative performance tuning
- Leveraged Gemini API, to improve accuracy on edge cases and optimize performance on larger text excerpts

AXIOM | *MapboxGL, Next.js, TypeScript, Three.js, Mapbox GL, Supabase*



- Built a full-stack AI platform that models grading schemes across Canadian high schools to normalize GPA comparisons
- Developed an interactive map-based interface with MapboxGL enabling users to explore and compare 2,000+ schools
- Deployed a multi-agent backend system to retrieve data, analyze rankings, and answer user queries with grounded data
- Built web-scraping and processing pipeline to extract structured school data from several sources for efficient querying

NewDen | *Python, React, JavaScript, spaCy, Gemini API*



- Built a full-stack NLP system for scam-detection, helping identify over 3,000 fraudulent housing listings in Ontario
- Extracted key structured entities and used them to improve scam detection beyond keyword-based methods
- Custom-trained an NER model (~1,000 samples per entity) to extract structured fields such as address and city name
- Leveraged Gemini API to aid processing speeds and enable comparisons of real-world housing data from any region